A Multi-Stage Artificial Intelligence Framework for Proactive Livestock Health Monitoring and Outbreak Surveillance

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***Abstract*—Management of livestock disease in small farms is frequently reactive: action begins only after clinical signs appear or an outbreak has already spread. Conventional precision-livestock technologies remain difficult to deploy at scale due to cost and infrastructure constraints. Existing research also tends to treat farm monitoring, image-based diagnosis, spatial analysis, and regional risk estimation as separate problems.**

We present the Animal Disease Reporting System (ADRS), an integrated decision-support architecture that links four stages: (i) farm-level health and environmental triage, (ii) protected image-based diagnosis, (iii) spatial-temporal outbreak analysis, and (iv) district-level outbreak-risk nowcasting. Only selected components have been quantitatively evaluated. These include object detection, single-label disease classification, clinical lesion segmentation, FMD spatial risk ranking, and concurrent FMD and LSD district-level nowcasting. Wellness triage, out-of-distribution screening, multi-label classification, and prognostic assessment remain design elements that have not yet been tested.

On the evaluated tasks, the system produced the following results. A ViT-B/16 classifier reached 88.78% accuracy (weighted F1 = 0.89; 95% Wilson CI 86.92–90.39%). A ResNet50 segmentation model achieved IoU = 0.8245 and F1 = 0.9028. The best FMD spatial-ranking model obtained a mean PR-AUC of 0.6379. Held-out 2024 FMD nowcasting yielded ROC-AUC = 0.8166 and PR-AUC = 0.6950. LSD nowcasting, however, showed unstable cross-year generalization (2024 ROC-AUC = 0.3383).

Taken together, the results indicate that the individual building blocks of ADRS are feasible and that the overall architecture is a useful organizing framework for decision support. They do not yet demonstrate validated end-to-end diagnostic or outbreak-control performance.

***Index Terms*—Precision livestock farming, livestock disease surveillance, deep learning, out-of-distribution detection, spatio-temporal clustering, outbreak-risk nowcasting.**

I. Introduction

A. Background and Agrarian Context

Livestock production supports food security, rural livelihoods and agricultural income in Sri Lanka. Roughly 1.65 million smallholder households depend on the sector, yet domestic dairy output still falls short of national demand [1], [2]. The introduction of higher-yield exotic and cross-bred cattle has raised productive potential, but health management in tropical conditions remains challenging [2]. Disease events impose direct production losses and wider economic costs; the recent outbreak of lumpy skin disease (LSD) in Ampara, for example, produced measurable farm-level impacts [3].

Surveillance that can detect abnormal states before they escalate into larger management problems is therefore valuable. In practice, however, observation is often periodic and depends on the availability of farmers or veterinary staff. Continuous assessment and the linkage of individual observations to larger spatial patterns remain limited.

B. Problem Statement and Surveillance Gaps

Four gaps are especially relevant for smallholder settings. First, precision livestock farming (PLF) enables continuous monitoring, yet cost and infrastructure constraints restrict deployment. Second, image-based diagnosis is usually framed as a closed-set classification task; this makes rejection of unfamiliar inputs difficult and prevents single-label outputs from representing concurrent conditions. Third, temporal alarm detection and spatial analysis are typically handled in isolation. Fourth, district-level veterinary records are sparse, zero-heavy, and irregularly reported, complicating risk estimation. These limitations motivate a workflow that connects farm monitoring, diagnostic assessment, spatial analysis, and regional risk estimation within a single architecture.

C. The Proposed ADRS Platform

ADRS links four complementary stages: (i) farm-level health and environmental triage, (ii) guarded image-based diagnosis, (iii) spatio-temporal outbreak analysis, and (iv) district-level outbreak-risk nowcasting. Information moves from individual observations toward broader surveillance while remaining under veterinary oversight.

Our contribution is system integration and component-level evaluation rather than a novel standalone diagnostic or epidemiological algorithm. The main contributions are:

1) an integrated ADRS architecture that connects farm monitoring, guarded diagnosis, spatio-temporal analysis, and district-level risk nowcasting;

2) an evaluated image-analysis pipeline for cattle and disease-region detection, single-label classification, and ResNet50-based lesion segmentation, together with specified but still unevaluated out-of-distribution and multi-label extensions;
3) a geospatial surveillance layer that combines event preprocessing and spatial risk ranking, with spatio-temporal clustering defined for future active-outbreak analysis; and
4) a district-month risk-assessment pipeline designed for sparse veterinary records and evaluated separately for foot-and-mouth disease (FMD) and LSD.

II. Related Work

Precision livestock farming combines sensing and computational analysis to recognize changes in animal condition earlier [4]. Smallholder deployment, however, is constrained by cost and infrastructure [5]. Automated body-condition scoring has been explored with both depth sensing and two-dimensional vision, although performance remains sensitive to farm conditions and image acquisition [6], [7]. Thermal stress can be quantified through the temperature–humidity index (THI) [8]. These studies motivate multimodal monitoring but do not yet provide a low-cost, integrated triage workflow suited to the target setting.

Computer vision has shown promise for livestock disease recognition, including LSD classification, although dataset diversity and external validation remain important limitations [9]. Closed-set classifiers do not explicitly reject unfamiliar inputs; out-of-distribution (OOD) detection supplies one such mechanism [10], while multi-label prediction can represent concurrent conditions.

At population scale, the World Organisation for Animal Health's WAHIS and the FAO's EMPRES-i+ systems provide disease reporting and early-warning functions [11], [12]. ST-DBSCAN can identify concentrations of observations under joint spatial and temporal constraints [13], yet clusters alone do not establish transmission pathways. Environmental and host-related variables supply additional disease-specific context [14]. Classical surveillance methods such as the Farrington algorithm account for trend, seasonality, and overdispersion [15], while zero-inflated negative-binomial models have been applied to sparse FMD counts [16]. Conformal prediction offers uncertainty sets, although temporal dependence and non-stationarity require careful validation [17].

These studies inform the task-specific model choices adopted in ADRS. The selection rationale is given in Section III-F; we make no claim of universal superiority.

III. Methodology and System Architecture

A. Overall ADRS Methodological Framework

ADRS operates at three scales: individual animal, local cluster, and district. Longitudinal animal and environmental observations feed a wellness-triage stage. When image evidence is available it enters a guarded diagnostic pipeline for integrity checking, disease classification, and lesion analysis. Accepted disease reports—containing disease identity, confidence, time, location, and quality metadata—are passed to the geospatial layer. In parallel, a district-month model combines surveillance history with environmental and spatial predictors.

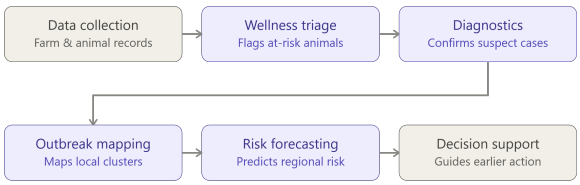


Fig. 1. ADRS data flow linking farm-level wellness triage, guarded diagnosis, spatio-temporal surveillance, and district-month outbreak-risk nowcasting. Solid boxes indicate evaluated components; dashed boxes indicate proposed elements that have not yet been quantitatively validated.

Throughout the paper we distinguish components that have been quantitatively evaluated from those that remain design proposals.

B. Multi-Modal On-Farm Wellness Triage

The proposed wellness component combines longitudinal physiological observations, environmental context, and mobile imagery. Daily milk yield and body weight form a persistent stream. A detected deviation or farmer concern opens a seven-day observation window that records additional feeding, water intake, and behavioral information.

A vision stage localizes the bovine region with YOLOv8 and extracts landmarks (hooks, pins, spine) for body-condition-score estimation. Environmental context is obtained from GPS-linked temperature and relative humidity, converted to the temperature–humidity index. A late-fusion classifier is intended to map visual BCS, the seven-day observation window, and thermal context into one of three states: Healthy, Heat Stress, or Unhealthy. Grad-CAM is proposed for visual attribution [18]. This component has not been quantitatively evaluated in the present study.

C. In-Field Diagnostic and Prognostic Assessment

The diagnostic pipeline comprises four stages.

- **Stage 1 – Input integrity.** Bovine verification is performed with YOLOv8s; a convolutional autoencoder is specified for out-of-distribution detection (unevaluated).
- **Stage 2 – Disease classification.** A ViT-B/16 model produces single-label predictions. Multi-label output remains an unevaluated extension.
- **Stage 3 – Lesion segmentation.** A ResNet50 encoder with a transposed-convolution decoder estimates lesion extent and burden on clinical field images (evaluated).
- **Stage 4 – Prognostic assessment.** A multilayer perceptron that combines lesion burden with anatomical cues (ear posture, ocular appearance) is specified but unevaluated.

Detection and classification are assessed with standard metrics; segmentation is assessed with intersection-over-union (IoU) and F1-score.

D. Spatio-Temporal Outbreak Analysis

Geolocated FMD and LSD observations are first cleaned: labels, coordinates, and timestamps are validated, duplicates are removed, and spatial precision is checked. Distance calculations use a projected coordinate system; geographic coordinates are retained only for display.

ST-DBSCAN is specified for disease-specific spatial and temporal neighborhoods [13]. Parameter choices should be guided by epidemiological knowledge, sensitivity analysis, and historical back-testing. Clusters indicate concentrations of reported events rather than proven transmission pathways. The present evaluation covers spatial risk ranking only; trajectory estimation, directional fields, and quality-weighted kernel-density surfaces remain unevaluated.

E. Concurrent District-Month Outbreak-Risk Classification and Nowcasting

The district-level component estimates concurrent outbreak risk across Sri Lanka's 25 districts. The processed FMD data set contains 2 400 district-month observations (2017–2024); the LSD data set contains 1 500 observations (2020–2024).

Features combine disease history with rainfall, meteorological variables, seasonal indicators, livestock density, teleconnection indices, and neighboring-district history. Precipitation is taken from CHIRPS; additional meteorological fields are derived from ERA5 and NASA POWER [19]–[21]. Seasonality is encoded with sine/cosine month terms and monsoon indicators. The FMD model also includes a previous-month outbreak indicator.

Scaling is fitted on training observations only; chronological separation is preserved throughout. FMD uses L2-regularized logistic regression; LSD uses Elastic-Net with Platt calibration [22], [23]. A Random Forest can supply advisory non-linear predictions [24]. Conformal uncertainty remains a design element. Feature contributions for the linear models are computed as

$$a_k = \beta_k \left(x_{k,i,m} - \bar{x}_k \right), \tag{1}$$

where a_k is the contribution of feature k, β_k its coefficient, and $\bar{x}_k$ the reference mean.

FMD is evaluated with a chronological held-out 2024 period. LSD is evaluated with walk-forward and leakage-corrected leave-one-year-out procedures. Because climate features include the target month, the results are concurrent risk nowcasts rather than long-lead forecasts.

F. Model Selection Rationale

Model choices reflect input characteristics, data availability, computational constraints, and the requirements of a veterinary decision-support setting. ViT-B/16 was selected for classification because self-attention captures relationships among image regions and transfer learning is practical for modest clinical data sets [25]. ResNet50 supplies a residual encoder that supports deep representation learning while remaining practical for lesion segmentation [26]. YOLOv8s offers a one-stage detection path and a compact variant suited to field-oriented inference [27]. Logistic regression provides an interpretable risk model; Elastic-Net is useful when environmental and spatial predictors are correlated [22]. Random Forest supplies a non-linear advisory model for interactions that a linear rule may miss [24]. These are pragmatic, task-specific choices, not claims of universal superiority.

G. Inter-Component Integration and Evaluation Design

Wellness deterioration can trigger diagnostic assessment. Accepted diagnostic evidence (disease identity, confidence, time, location) can form an epidemiological event. District-level risk estimates can contextualize local spatial patterns. These interfaces are treated as system integrations rather than validated end-to-end performance. All outputs are intended to support, not replace, professional veterinary diagnosis or authorized disease-control decisions.

IV. Experimental Results

Table I summarizes the quantitative results obtained for the evaluated components. Wellness triage, OOD screening, prognostic inference, and multi-label classification have not been quantitatively evaluated.

TABLE I
Summary of Quantitative Results

Component	Metric	Value
YOLOv8s detection	mAP@0.5	0.849
ViT-B/16 classification	Accuracy / weighted F1	88.78% / 0.89
ResNet50 segmentation	IoU / F1	0.8245 / 0.9028
FMD spatial ranking	Mean PR-AUC	0.6379
FMD nowcasting (2024 held-out)	ROC-AUC / PR-AUC	0.8166 / 0.6950
LSD nowcasting (purged LOYO mean)	ROC-AUC / PR-AUC	0.6291 / 0.2441

A. Clinical Diagnostics and Symptom Segmentation

The YOLOv8s detector achieved an mAP@0.5 of 0.849 on a held-out test set of 942 images containing 1 278 annotated instances. Mastitis showed the lowest recall (0.663).

Training and validation losses decreased together across 110 epochs without sustained divergence. Validation precision and recall rose from below 0.55 to approximately 0.80; mAP@0.5 stabilized near 0.81 and mAP@0.5:0.95 near 0.53. These validation figures are not directly comparable with the held-out test result reported in Table I.

The ViT-B/16 classifier achieved 88.78% accuracy (1 131 correct predictions out of 1 274 samples) with a 95% Wilson confidence interval of 86.92–90.39%. Weighted F1 was 0.89. Performance was strongest for FMD (F1 = 0.95) and weakest for mastitis (F1 = 0.78).

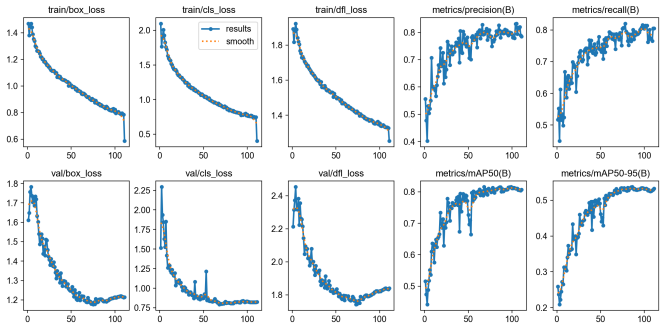


Fig. 2. Training and validation curves for the YOLOv8s detector over 110 epochs.

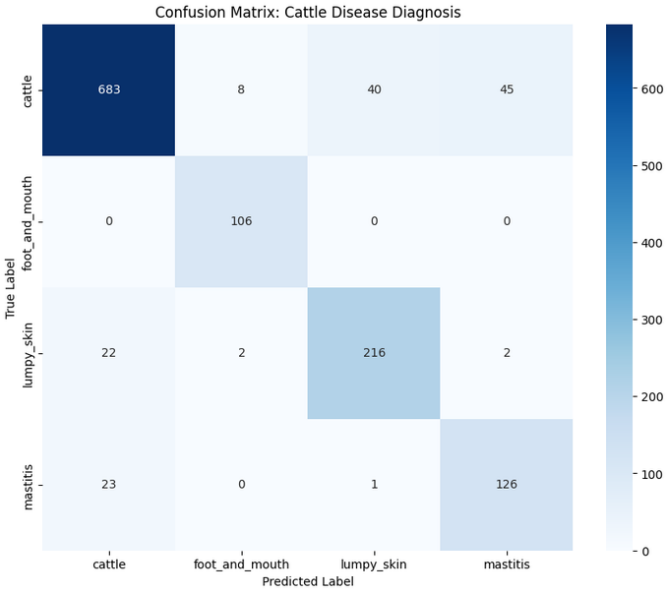


Fig. 3. Confusion matrix for the ViT-B/16 single-label disease classifier.

The confusion matrix reveals few direct confusions among the named disease classes. The dominant errors occur between healthy and disease categories: 40 healthy images were classified as LSD and 45 as mastitis, while 22 LSD and 23 mastitis images were classified as healthy. This pattern suggests that visually subtle presentations remain harder to separate from healthy animals than established disease categories are from one another.

The ResNet50-based segmentation model achieved an IoU of 0.8245 and an F1-score of 0.9028 on clinical field images, indicating good delineation of the annotated disease regions in the evaluated data.

B. Geospatial Risk Ranking

Thirty candidate FMD spatial-ranking models were evaluated across chronological folds. The simple distance-only exponential model with a 25 km scale achieved a mean PR-AUC of 0.6379, outperforming the best Gradient Boosted Trees candidate (0.5916). A Gaussian distance function at the same scale yielded 0.6346. These results suggest that spatial scale may matter more than the precise form of the decay function in the data examined here.

The reported values reflect relative ranking performance, not calibrated transmission probabilities. Because the same folds were used for both model selection and final reporting, the estimate may be optimistic; a fully separated final evaluation period would provide stronger evidence of generalization.

C. Concurrent District-Month Risk Nowcasting

The FMD logistic-regression model achieved ROC-AUC = 0.8166 and PR-AUC = 0.6950 on the held-out 2024 period. No deployment threshold is recommended from these threshold-independent metrics.

The LSD Elastic-Net model achieved mean ROC-AUC = 0.6291 and PR-AUC = 0.2441 under purged leave-one-year-out evaluation. The 2024 fold produced ROC-AUC = 0.3383—below chance. We treat this result as an instability signal that warrants careful checks of label encoding, fold composition, positive-class counts, reporting shifts, and feature construction.

V. Discussion and Limitations

The results demonstrate the feasibility of several core ADRS components, yet they fall short of establishing complete end-to-end surveillance performance. The strongest evidence lies in the image-analysis pipeline and in FMD nowcasting. Wellness triage, OOD screening, prognostic inference, and multi-label classification remain unevaluated.

The ViT-B/16 accuracy of 88.78% should be read together with its class-wise results and confidence interval. The ResNet50 segmentation figures are encouraging on the available clinical images, but broader testing across farms, regions, disease presentations, and acquisition conditions is still required. The YOLOv8s result is likewise a component-level finding obtained under the evaluated field-oriented conditions.

A simple epidemiologically motivated distance function produced the highest PR-AUC among the spatial-ranking candidates, indicating that useful spatial structure can be captured without complex models. Model-selection reuse, however, may inflate the reported estimate. The below-chance LSD result for 2024 further underscores the necessity of disease-specific temporal validation; the score could reflect labeling, sampling, reporting, or distribution-shift issues rather than a true inverse relationship.

PR-AUC values should always be interpreted in light of positive-class prevalence, because the random-ranking baseline depends on prevalence. The present manuscript does not supply sufficient prevalence information for such a comparison. Uncertainty is currently quantified only for the ViT accuracy; stronger validation should report fold-wise dispersion and calibration metrics for the ranking and nowcasting tasks.

Because the district models incorporate same-month climate observations, they constitute concurrent risk nowcasts rather than long-lead forecasts. Spatial clusters are likewise treated as concentrations of reported events, not as causal transmission pathways.

Ethics and Data Availability

The study uses animal-related imagery and farm records that may contain location information. All data collection, use, and consent procedures complied with the responsible institutional and regional review requirements.

For operational deployment, record-level GPS coordinates should remain access-controlled. Broader dissemination should rely on aggregated spatial representations or administrative advisory zones.

Data Availability: The livestock disease records and related surveillance data used in this study were obtained from the Department of Animal Production and Health (DAPH), Sri Lanka. Due to institutional data-sharing agreements, privacy considerations, and the presence of location-sensitive farm information, the raw records cannot be made publicly available. Aggregated or de-identified versions of the processed features, together with the analysis code, may be shared by the authors upon reasonable request and subject to approval by DAPH. Clinical images used for the vision components were collected under the same institutional arrangements and are likewise restricted.

VI. Conclusion

ADRS connects farm-level monitoring, image-based diagnosis, spatial outbreak analysis, and district-level disease-risk nowcasting into a single decision-support architecture. The evaluated detection, classification, segmentation, spatial-ranking, and FMD nowcasting components provide encouraging component-level evidence. Unstable LSD generalization and several still-unevaluated stages prevent any claim of fully validated end-to-end performance.

The principal contribution is therefore the integration architecture itself, together with an evidence-aware evaluation strategy that clearly separates implemented components from proposed extensions.

Future work should prioritize:

- reporting positive-class prevalence for all ranking and nowcasting tasks,
- broader uncertainty quantification and calibration analysis,
- systematic investigation of the below-chance LSD 2024 fold,
- fully separated spatial model-selection and final-evaluation folds, and
- an end-to-end field trace that exercises the interfaces between ADRS stages under realistic operating conditions.

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